

40.5 Working correlation

Theorem 40.4.1 says the working correlation may be wrong without costing consistency. It does not say the choice is free. A wrong one costs *efficiency*, changes the small-sample behaviour of the sandwich, and — because it changes the estimate — can be turned into a diagnostic for the mean model.

40.5.1 The catalogue

The structures in use are those of Definition 33.4.1, written as correlation matrices and estimated by moments rather than by likelihood. For a cluster with n_i observations at occasions $j = 1, \dots, n_i$, and Pearson residuals $r_{ij} = (y_{ij} - \hat{\mu}_{ij}) / \sqrt{V(\hat{\mu}_{ij})}$ with $\hat{\phi} = \sum_{ij} r_{ij}^2 / (n - p)$:

Structure	$(\mathbf{R}_i)_{jk}, j \neq k$	Parameters
independence	0	0
exchangeable	α	1
first-order autoregressive	$\alpha^{ j-k }$	1
q -dependent	$\alpha_{ j-k }$ if $ j - k \leq q$, else 0	q
unstructured	α_{jk}	$n(n - 1)/2$

The parameters are estimated by matching averages of residual products. Writing $N_p = \sum_i \binom{n_i}{2}$ for the within-cluster pairs and $N_a = \sum_i (n_i - 1)$ for the adjacent ones,

$$\hat{\alpha}_{\text{exch}} = \frac{1}{\hat{\phi}(N_p - p)} \sum_i \sum_{j < k} r_{ij} r_{ik}, \quad \hat{\alpha}_{\text{ar}} = \frac{1}{\hat{\phi}(N_a - p)} \sum_i \sum_{j < n_i} r_{ij} r_{i,j+1},$$

and $\hat{\alpha}_{jk} = \sum_i r_{ij} r_{ik} / (\hat{\phi} m)$, rescaled to have unit diagonal, for the unstructured case; the q -dependent estimates are lag-by-lag averages of the same kind.

The design narrows the list at once. The unstructured matrix needs a common set of occasions across clusters and costs $n(n - 1)/2$ parameters; the autoregressive structure needs an ordering, so it is for longitudinal data and not for pupils in a school; the exchangeable structure is the one a random intercept induces (Proposition 40.2.1), and is the default for clustered data with no natural order. And every estimated $\mathbf{R}_i$ must be a valid correlation matrix, which the moment estimates above do not guarantee — Example (Four options per traveller) showed the exchangeable estimate landing almost exactly on the boundary.

```
import numpy as np

KINDS = ["independence", "exchangeable", "ar1", "unstructured"]

def correlation(kind, r, p, phi):
    """Moment estimates of the working correlation from Pearson residuals r (m by n)
    ."""
    m, n = r.shape
    if kind == "independence":
        return np.eye(n)
    if kind == "exchangeable":
        off = (np.sum(r.sum(1) ** 2) - np.sum(r**2)) / 2
        a = off / (phi * (m * n * (n - 1) / 2 - p))
```

```

        return np.eye(n) + a * (1 - np.eye(n))
    if kind == "ar1":
        a = np.sum(r[:, :-1] * r[:, 1:]) / (phi * (m * (n - 1) - p))
        return a ** np.abs(np.arange(n)[:, None] - np.arange(n))
    if kind == "unstructured":
        R = (r.T @ r) / (phi * m)
        return R / np.sqrt(np.outer(np.diag(R), np.diag(R)))
    raise ValueError(kind)

def gee(y, X, kind, family="poisson", maxit=100):
    m, n, p = X.shape
    beta = np.zeros(p)
    beta[0] = np.log(max(y.mean(), 1e-3)) if family != "binomial" else 0.0
    for _ in range(maxit):
        eta = X @ beta
        mu = np.exp(eta)
        dmu, V = (mu, mu) if family == "poisson" else (mu, mu**2)
        r = (y - mu) / np.sqrt(V)
        phi = np.sum(r**2) / (m * n - p)
        R = correlation(kind, r, p, phi)
        Rinv = np.linalg.pinv(R)
        SX = (dmu / np.sqrt(V))[..., None] * X
        A = np.einsum("mjp,jk,mkq->pq", SX, Rinv, SX)
        step = np.linalg.solve(A, np.einsum("mjp,jk,mk->p", SX, Rinv, r))
        beta = beta + step
        if np.max(np.abs(step)) < 1e-10:
            break
    u = np.einsum("mjp,jk,mk->mp", SX, Rinv, r)
    Ainv = np.linalg.inv(A)
    return beta, Ainv @ (u.T @ u) @ Ainv, phi, R

```

[Run this code online](#)

40.5.2 Efficiency, and where it comes from

Proposition 40.5.1 — Choosing the working correlation

In the setting of Theorem 40.4.1:

- Efficiency.** The asymptotic covariance $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$ of Eq. 40.4.5 is minimized, in the nonnegative-definite ordering and over all unbiased estimating functions linear in $\mathbf{Y}_i - \boldsymbol{\mu}_i$, by $\mathbf{V}_i = \boldsymbol{\Sigma}_i^0$ (Exercise (C1)). Any other working covariance costs efficiency, but the cost depends on the covariate: it is substantial for a covariate that varies *within* clusters and negligible for one that is constant within a cluster, for which the independence choice is in fact hard to beat.
- Small samples.** The sandwich Eq. 40.4.6 is biased downwards by a term of order p/m , so Wald tests built on it reject too often when m is small; the bias-corrected forms of Section 40.4 and a t_{m-p} reference repair most of it. The bias grows with the number of parameters in $\mathbf{R}_i$, because $\hat{\boldsymbol{\alpha}}$ is then noisier and condition (E5) of Theorem 40.4.1 is stretched.
- QIC.** Since there is no likelihood, AIC is unavailable. Pan's **quasi-likelihood information criterion** replaces the log-likelihood by the quasi-likelihood $Q(\boldsymbol{\mu}; \mathbf{y}) =$

$\sum_{ij} \int_{y_{ij}}^{\mu_{ij}} (y_{ij} - t) / \{\phi V(t)\} dt$ of Definition 38.2.1 computed under the *independence* working correlation at the fitted $\hat{\boldsymbol{\mu}}$, with ϕ the Pearson estimate of Proposition 38.3.2 held fixed across the models compared, and replaces the parameter count by a sandwich-based trace:

$$\text{QIC} = -2Q(\hat{\boldsymbol{\mu}}; \mathbf{y}) + 2 \text{tr}(\hat{\boldsymbol{\Omega}}_I \hat{\mathbf{U}}_R), \quad (40.5.1)$$

where $\hat{\boldsymbol{\Omega}}_I = \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{A}}_i^{-1} \hat{\mathbf{D}}_i / \phi$ is the independence-model information and $\hat{\mathbf{U}}_R$ is Eq. 40.4.6. If the working model is correct the trace is p and Eq. 40.5.1 is the Akaike form Definition 29.2.2. It is a sound criterion for choosing the *mean model*, and a weakly justified one for choosing $\mathbf{R}_i$.

Proof.

- (a) is Exercise (C1); the covariate-dependence is illustrated in Example (How much efficiency is at stake) and explained after it. (b) is the cluster version of the bias calculation in Exercise (C2), applied to $\hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i$ in place of $\mathbf{X}_g^\top \hat{\boldsymbol{\epsilon}}_g$. (c) is a definition; the claim that the trace equals p when the working model is correct is Theorem 40.4.1(d), which gives $\hat{\mathbf{U}}_R \rightarrow \mathbf{A}^{-1}$, together with $\boldsymbol{\Omega}_I = \mathbf{A}$ when $\mathbf{R}_i = \mathbf{I}$ is correct. ■

Remark (Why QIC is weak as a chooser of correlation). Eq. 40.5.1 measures fit by a quasi-likelihood that ignores the correlation altogether, and penalizes by a trace that is smallest when the sandwich is closest to the model-based covariance. Changing $\mathbf{R}_i$ therefore moves QIC through the penalty almost alone, and rewards whichever structure makes the sandwich small — not whichever is right. Pan (2001) proposed it primarily for the mean model, and Hin and Wang (2009) give a variant (CIC) that drops the fit term for correlation selection. Neither has the standing AIC has for likelihoods.

Example. (How much efficiency is at stake)

Simulate 60 clusters of 6 Poisson counts with a log link, one covariate varying within the cluster and one constant within it, and generate the dependence in two ways: a shared random intercept, which makes the true correlation exchangeable, and a latent Gaussian AR(1) process with $\rho = 0.8$. Fit by GEE with each of the four working correlations and record the sampling variance of each coefficient over 400 replications.

For the **within-cluster** covariate under exchangeable truth, the efficiency of the exchangeable working correlation relative to independence is 1.789; AR(1) and unstructured reach 1.583 and 1.547. Under AR(1) truth the figures are 1.282 for AR(1), 1.183 for exchangeable and 1.160 for unstructured. Matching the truth helps, mismatching it still helps, and the gain is worth having without being dramatic.

For the **cluster-level** covariate the ranking reverses: under exchangeable truth the relative efficiencies are 0.889, 0.938 and 0.834, all below one, so independence is best of the four. The reason is Exercise (B1): a covariate constant within a cluster is estimated from cluster means, on which an exchangeable reweighting has no effect at all, so all a nontrivial $\mathbf{R}_i$ contributes is the noise in $\hat{\boldsymbol{\alpha}}$.

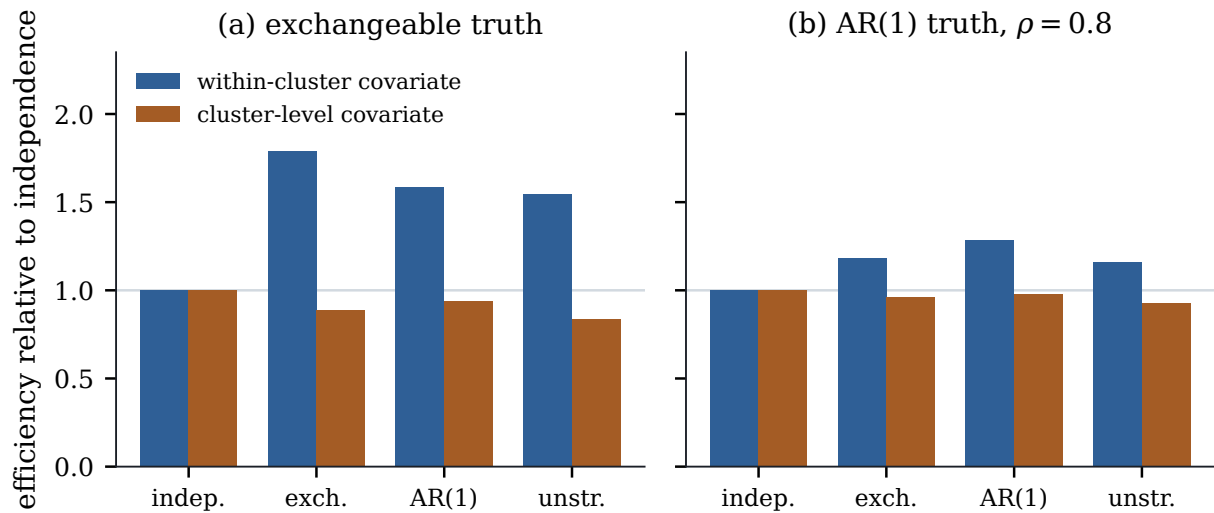


Figure 40.5.1: Efficiency of four working correlations relative to the independence choice, for a within-cluster and a cluster-level covariate, over 400 replications with $m = 60$ clusters of $n_i = 6$ Poisson counts. (a) The true dependence is exchangeable (a shared random intercept). (b) The true dependence is a latent AR(1) process with $\rho = 0.8$. Bars above one mean more efficient than independence.

40.5.3 A diagnostic hiding in the efficiency

By Theorem 40.4.1(a), every working correlation estimates the *same* β^0 , provided the mean model Eq. 40.4.1 is right. Two estimates of the same quantity differing by more than their sampling error therefore accuse the mean model, not the correlation. This is the Hausman idea, and it has real diagnostic value because the independence and exchangeable estimates weight within- and between-cluster variation differently: independence is dominated by between-cluster comparisons when clusters are large, and a nontrivial $\mathbf{R}_i$ shifts weight towards within-cluster ones.

Example. (Grunfeld's panel by GEE)

Fit the marginal version of Example (Grunfeld's firms, again) — gamma variance function $V(\mu) = \mu^2$, log link, investment on log value and log capital — to the 11 firms by GEE. The AR(1) working correlation estimates a lag-one correlation of 0.894, as one expects of annual investment figures. Yet QIC, with the dispersion held at the independence estimate of the larger model, ranks the four structures

Structure	QIC, value and capital	QIC, value only
independence	8450.1	8657.6
exchangeable	8524.2	8647.4
AR(1)	8574.1	8657.8
unstructured	8754.7	8704.1

and prefers independence. As a comparison of *mean* models the table is informative: at three of the four structures the model with capital wins, by between about 80 and 210 QIC units, and that is the comparison QIC was designed for. Under the unstructured correlation the ranking reverses, which with 11 clusters and a 20×20 matrix to estimate is a comment on the structure rather than on the model. As a comparison of structures the column mostly reports that the sandwich is smallest under independence — the remark above in numerical form.

The estimates themselves are the more interesting output. The independence fit gives $\hat{\beta}_{\text{value}} = 0.8421$ and $\hat{\beta}_{\text{capital}} = 0.3386$; the exchangeable fit gives 0.7214 and 0.2113. The gap in the first coefficient is more than its sandwich standard error of 0.1030, and the two gaps point the same way. By Theorem 40.4.1(a) that accuses the mean model, and two standard explanations fit equally well: firm effects correlated with the covariates, as Example (Grunfeld's firms, again) found, or feedback from investment to the following years' market value, which violates the full-covariate conditional mean condition and makes the exchangeable fit — not the independence one — the inconsistent member of the pair. Either way the remedy is to separate the two sources of variation. Adding the firm means of log value and log capital, as Exercise (C1) asks, makes the independence and exchangeable estimates agree to machine precision, at within-firm slopes 0.6142 and 0.2298, both well below the independence figures above.

```
import statsmodels.api as sm

d = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
m, n = d["firm"].nunique(), 20
y = d["invest"].to_numpy().reshape(m, n)
lv = np.log(d["value"].to_numpy()).reshape(m, n)
lc = np.log(d["capital"].to_numpy()).reshape(m, n)

def qic(y, X, kind, phi):
    """Pan's QIC: the quasi-likelihood of the independence model at the fitted mean,
    plus a
    penalty built from the sandwich. For  $V(\mu) = \mu^2$  the quasi-likelihood is  $-y/\mu - \log \mu$ .
    The dispersion  $\phi$  is held fixed across the models compared, as in Mallows's  $C_p$ .
    """
    beta, robust, _, _ = gee(y, X, kind, family="gamma")
    mu = np.exp(X @ beta)
    Q = np.sum(-y / mu - np.log(mu)) / phi
    omega = np.einsum("mjp,mjq->pq", X, X) / phi      # independence information, log
    link
    return -2 * Q + 2 * np.trace(omega @ robust)

X_full = np.stack([np.ones((m, n)), lv, lc], axis=2)
X_small = np.stack([np.ones((m, n)), lv], axis=2)
phi_ref = gee(y, X_full, "independence", family="gamma")[2]      # from the largest
model

for name, XX in [("value + capital", X_full), ("value only", X_small)]:
    for kind in KINDS:
        print(f"{name:16s} {kind:14s} QIC {qic(y, XX, kind, phi_ref):10.2f}")
```

[Run this code online](#)

40.5.4 Practical advice

- **Count the clusters first.** Everything here that uses a sandwich needs m large. Below about forty clusters use a bias correction and a t reference; below about fifteen, prefer a likelihood-based model with a covariance structure chosen by REML (Proposition 33.4.1) and accept the assumptions.
- **Match the structure to the design, not to the data.** Exchangeable for unordered clusters, autoregressive for equally spaced longitudinal data, unstructured only when the occasions are common to all clusters and m is large.
- **Use independence when the covariates are time-varying, or the covariate of interest is cluster-level.** In the first case a non-diagonal $\mathbf{R}_i$ needs the full-covariate conditional mean condition of Section 40.4, which feedback destroys; in the second it loses nothing (Example (How much efficiency is at stake)). Independence needs neither, estimates nothing extra, and cannot produce an invalid $\mathbf{R}_i$.
- **Compare structures to check the mean model.** A large difference between the independence and exchangeable estimates is a warning about $\mathbf{X}$, not about $\mathbf{R}_i$.
- **Say which covariance you report,** and never put a GEE coefficient and a GLMM coefficient in the same table as if they estimated the same thing. The first costs coverage (Example (Coverage when the working correlation is wrong)); the second is the commonest error in this literature.

Remark (Beyond correlations). For binary responses a correlation is an awkward measure of dependence, because the Fréchet bounds on the joint cell probability confine it to an interval that depends on the two means, while a pairwise odds ratio is free in $(0, \infty)$ whatever the means are. **Alternating logistic regressions**, due to Carey, Zeger and Diggle (1993), exploit this: they replace $\mathbf{R}_i$ by a model for the pairwise odds ratios and alternate between the mean equation Eq. 40.4.3 and a logistic regression for the association, whose parameters are unconstrained and interpretable in their own right.

40.5.5 Exercises

40.5.5.1 A. Check your understanding

Exercise A1.

For each design, name the working correlation you would use and say why: (a) pupils within 300 schools, class size varying; (b) five annual measurements on each of 500 patients; (c) left and right eyes of 80 patients; (d) irregularly timed clinic visits, between two and eleven per patient, 1200 patients.

Exercise A2.

In Example (Grunfeld's panel by GEE), QIC prefers independence among the working correlations but the AR(1) parameter is estimated at 0.894. Are these two statements in conflict? Explain what each of them is measuring.

40.5.5.2 B. Practice

Exercise B1.

Show that the exchangeable correlation matrix $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{1}\mathbf{1}^\top$ of size n has eigenvalues $1 + (n - 1)\alpha$ (once) and $1 - \alpha$ ($n - 1$ times), so that it is positive definite exactly when $-1/(n - 1) < \alpha < 1$. Verify the value $1 + 3\hat{\alpha} = 0.0956$ quoted in Example (Four options per traveller) from $\hat{\alpha} = -0.3015$, and say what a practitioner should do when a moment estimate falls outside the range.

Solution. $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{1}\mathbf{1}^\top$; $\mathbf{1}$ is an eigenvector with eigenvalue $(1 - \alpha) + n\alpha = 1 + (n - 1)\alpha$, and any vector orthogonal to $\mathbf{1}$ has eigenvalue $1 - \alpha$. Positivity of both gives the range. With $\hat{\alpha} = -0.3015$ and $n = 4$, $1 + 3(-0.3015) = 1 - 0.9045 = 0.0955$, which matches the quoted 0.0956 to rounding. Outside the range the working covariance is not a covariance at all and the update Eq. 40.4.4 can diverge; the usual remedies are to shrink $\hat{\alpha}$ towards zero, to project $\mathbf{R}$ onto the nearest correlation matrix, or — better — to use the independence structure, which is never invalid.

Exercise B2.

Formalize the diagnostic of [Example \(Grunfeld's panel by GEE\)](#). Let $\hat{\boldsymbol{\beta}}_I$ and $\hat{\boldsymbol{\beta}}_E$ be the independence and exchangeable GEE estimates. Explain why $\hat{\boldsymbol{\beta}}_I - \hat{\boldsymbol{\beta}}_E$ has mean zero to first order when the mean model is correct, and why its covariance is *not* the difference of the two covariances in general. What resampling scheme would give a usable reference distribution?

Solution. Both estimates satisfy Eq. 40.4.5 with the same $\boldsymbol{\beta}^0$, so their difference is $o_p(1)$ and, after scaling, asymptotically normal with mean zero. The classical Hausman simplification $\text{Cov}(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2) = \text{Cov}(\hat{\boldsymbol{\beta}}_2) - \text{Cov}(\hat{\boldsymbol{\beta}}_1)$ needs one of the two to be *efficient*, which neither is here, so the covariance must be obtained from the joint asymptotic distribution of the two estimating functions or by resampling. The cluster bootstrap of Exercise (C1) — resample whole clusters with replacement and refit both — gives the reference distribution directly.